Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Integrate raw Olympic datasets (medals.csv, athlete_events.csv, athlete.csv) from external CSV repositories into a centralized analysis environment.	High
2	Data Processing	Perform data cleaning operations, handle missing values (imputation or filtering), remove duplicate records, and standardize text formatting across all loaded datasets.	High
3	Business Analysis	Calculate core Olympic KPIs, including total medal counts (Gold, Silver, Bronze), host country performance metrics, and athlete demographic success rates.	High
4	Business Insights & Analytical Outputs	Provide dynamic interactive dashboards, comparative visualizations, and automated summary reports showcasing historical performance trends by nation and sport.	High
5	User Interaction & Analysis	Allow users to apply filters (e.g., Year range, specific Sport, Country/NOC) for multidimensional exploration and drill-down analysis within the visualizations.	High
6	Data Update & Reporting	Schedule regular data refresh processes for new Olympic games and generate downloadable performance summary reports for stakeholders.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description

NFR-1	**Usability**	The solution must provide an intuitive user interface with interactive dashboards and clear visualizations that are easy for non-technical sports analysts to navigate with minimal training.
NFR-2	**Security**	Access to raw data and analysis results should be restricted based on user roles; secure data handling pipelines must prevent unauthorized access to the underlying databases.
NFR-3	**Reliability**	The system must ensure high data accuracy and dashboard consistency, with automated alerts in place to notify users in case of data integration or refresh failures.
NFR-4	**Performance**	Large dataset queries and interactive dashboard filters must execute quickly, rendering visualizations within 2-3 seconds to maintain a smooth user experience.
NFR-5	**Availability**	The interactive dashboard solution should have 99% uptime, ensuring reliable access for users during crucial sports analysis and research periods.
NFR-6	**Scalability**	The architecture must be scalable to accommodate the addition of future Olympic games data and handle increasing concurrent user access without significant degradation in performance.